

ASANDA KHUMALO

Product-Focused Software Engineer (Full-Stack | Mobile & Web)

as.nkab01@gmail.com | +27 66 132 2024 | Durban

GitHub: github.com/NDOCY | LinkedIn: <https://www.linkedin.com/in/asanda-khumalo-7a4a76297?>
| Portfolio: <https://ndocy.github.io/portfolio/>

PROFESSIONAL SUMMARY

Product-focused software engineer with hands-on experience building scalable mobile, backend and web systems across retail, agriculture and operational management domains.

Skilled in Flutter, ASP.NET MVC, SQL Server, Firebase and Supabase, with experience designing real-world workflows, relational systems and production-style architectures.

Experienced working across multiple technologies including Node.js and Python desktop systems, with a strong focus on execution, usability and shipping complete applications from concept to deployment.

TECHNICAL SKILLS

Languages:

- C#, Dart, Javascript, SQL, Python

Frameworks & Platforms:

- ASP.NET MVC (Framework), Entity Framework, Flutter, Firebase, Supabase, Android Studio, Node.js, Flask, Tkinter

Databases & Cloud:

- SQL Server (SSMS, Azure Data Studio), PostgreSQL (Supabase), Firebase, Microsoft Azure

Tools & Concepts:

- Git & GitHub, REST APIs, Authentication & Authorization, MVC Architecture, Relational Data Modelling, Role-Based Systems

PROJECT EXPERIENCE

Mobi Suite – Mobile POS & Inventory Management Platform

Founder & Software Developer

Tech: Flutter, Supabase, Firebase

- Designed and built a mobile point-of-sale system for small and medium retailers.
- Implemented inventory management, product categorization, and sales tracking.
- Designed transactional database structure using sales and sale_items to record every checkout and item price.
- Built the system to operate without expensive hardware, with support for optional peripherals.
- Focused on scalability, clean UI, and real-world retail workflows.

FarmTrack Pro – Smart Farm Management & E-Commerce System

Full-Stack Developer

Tech: C#, ASP.NET MVC, Entity Framework, SQL Server, Azure, Bootstrap, JavaScript

- Built a comprehensive web-based farm management system covering operations, workforce, livestock, inventory, crops, and sales.

- Implemented job applications and task management with recurring schedules, progress tracking, and department-based assignments.
- Designed inventory management with usage logs, restocking thresholds, and supplier-linked replenishment logic
- Developed livestock management including health records, weight tracking, vaccination schedules, and reproduction methods
- Implemented crop & plot lifecycle tracking with dashboards for planting, fertilization, pest control, irrigation, and harvesting
- Deployed academic environment using Azure App Service and Azure SQL Database

Westend Motors – Automotive Dealership Management System

Software Developer

Tech: C#, ASP.NET MVC, Entity Framework, SQL Server, Azure Blob Storage

- Developed a full-featured dealership management platform for vehicle sales, trade-ins, and servicing
- Implemented vehicle inventory management with specifications, pricing, images, and availability states
- Built a trade-in workflow covering submission, evaluation, inspection scheduling, approval, and inventory conversion
- Developed customer portal for browsing vehicles, booking services, and tracking service history
- Implemented appointment and service management with technician assignment and status tracking
- Designed role-based dashboards for Admin, Sales Staff, and Customers
- Integrated Azure Blob Storage for vehicle image uploads and management

Facility Resource Mapper — Resource Discovery & Navigation Platform

Software Developer

Tech Stack: Flutter, Firebase, Maps, Mobile Development

- Developed a cross-platform resource discovery system adaptable for campuses, malls, hospitals and large facilities.
- Built searchable resource mapping and navigation workflows using Firebase and Flutter.

- Focused on intuitive mobile UX, accessibility and scalable facility-based architecture.

Accounting System — Desktop Financial Management Platform

Software Developer

Tech Stack: Python, Tkinter, Desktop Development

- Built a desktop accounting and reporting platform for business workflow management.
- Implemented transaction tracking, financial reporting and workflow automation features.
- Developed desktop-focused interfaces and operational management functionality using Python and Tkinter.

Elestia — Backend & Platform Architecture Exploration

Software Developer

Tech Stack: Node.js, JavaScript, Backend Systems

- Explored scalable backend architecture patterns using Node.js and JavaScript.
- Developed foundational platform systems focused on modern backend workflows and API structures.
- Gained practical exposure to server-side development and scalable web technologies.

EDUCATION

Diploma in ICT: Application Development

Durban University of Technology

Key Areas:

- C# & ASP.NET MVC using Microsoft Visual Studio.
 - SQL development using SSMS and Azure Data Studio.
 - Cloud computing and deployments using Microsoft Azure.
 - Java-based Android development using Android Studio.
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ADDITIONAL

- Additional experience contributing to collaborative software projects involving authentication systems, backend integrations and full-stack application features across React, Flask and Python-based applications.
 - Strong self-taught experience in Flutter and product architecture beyond formal academic curriculum.
 - Comfortable working across mobile, backend and web platforms.
 - Strong product-thinking mind-set with focus on solving operational and workflow problems through software systems.
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